



MKSS's

Cummins College of Engineering for Women

Department of Electronics & Telecommunication

S.Y. B.Tech. (E&TC) 2025-2026 (SEM-I)

Open Ended assignment

Subject: Digital Electronics Lab (23PCEC303L)

Title : - Temperature-Controlled Fan System

Submitted by: Batch B1 Group No:2

CNUM: 1.UEC2024206

2.UEC2024209

3.UEC2024214

4.UEC2024215

Performance Date: 3-11-25

Problem Statement and Objective: -

In many environments—such as homes, offices, and industrial settings—maintaining optimal temperature is crucial for comfort, safety, and energy efficiency. Traditional fan systems require manual control, which can be inefficient and inconvenient. There is a need for an automated

solution that adjusts fan speed based on ambient temperature to ensure consistent cooling and reduce energy consumption.

To design and implement a Temperature-Controlled Fan System that:

- Automatically adjusts fan speed based on real-time temperature readings.
- Employs a microcontroller (e.g., Arduino, 8051, or FPGA with Verilog HDL) to process sensor data and control fan speed via PWM (Pulse Width Modulation).
- Enhances energy efficiency by reducing unnecessary fan usage.
- Provides a scalable foundation for smart cooling systems in homes, vehicles, and industrial setups.

Proposed Solution : -

To automate fan control based on ambient temperature, we implemented a temperature-controlled fan system using Verilog HDL.

The design integrates a temperature decoder and a PWM generator, both modeled behaviorally for clarity and synthesis efficiency. The system receives an 8-bit digital temperature input, categorizes it into four speed levels, and dynamically adjusts the fan's duty cycle using PWM signals. This modular approach ensures scalable control logic, energy-efficient operation, and real-time responsiveness. The solution is ideal for smart cooling applications in embedded systems, offering both reliability and adaptability.

Implementation details : -

a)Code:

Testbench for simulation

```
`timescale 1ns / 1ps
```

```

module tb_temp_controlled_fan;

    reg clk;
    reg [7:0] temp;
    wire fan;
    wire [1:0] speed_level;

    // Instantiate the DUT (Device Under Test)
    temp_controlled_fan uut (
        .clk(clk),
        .temp(temp),
        .fan(fan),
        .speed_level(speed_level)
    );

    // Generate clock: 20ns period = 50 MHz
    initial clk = 0;
    always #10 clk = ~clk;

    // Apply different temperature values
    initial begin
        $display("Time\tTemp\tFan\tSpeed");
        $monitor("%0t\t%d\t%b\t%b", $time, temp, fan, speed_level);

        temp = 8'd20; #2000; // Off
        temp = 8'd35; #2000; // Low
        temp = 8'd55; #2000; // Medium
        temp = 8'd80; #2000; // High
        temp = 8'd25; #2000; // Off again

        $stop;
    end
endmodule

```

endmodule
Verilog code

```

module temp_controlled_fan (
    input wire clk,           // Clock input (e.g., 50 MHz)
    input wire [7:0] temp,    // 8-bit temperature input
    output reg fan,           // PWM output to fan
    output reg [1:0] speed_level // 00=Off, 01=Low, 10=Med, 11=High
);

```

```

    reg [7:0] pwm_counter = 0;
    reg [7:0] duty_cycle = 0;

```

```

    // Determine duty cycle based on temperature
    always @(posedge clk) begin
        if (temp < 8'd30) begin
            duty_cycle <= 8'd0;    // Fan off
            speed_level <= 2'b00;
        end
        else if (temp < 8'd50) begin
            duty_cycle <= 8'd85;   // Low speed
            speed_level <= 2'b01;
        end
        else if (temp < 8'd70) begin
            duty_cycle <= 8'd170;  // Medium speed
            speed_level <= 2'b10;
        end
        else begin
            duty_cycle <= 8'd255;  // High speed
            speed_level <= 2'b11;
        end
    end
end

```

```

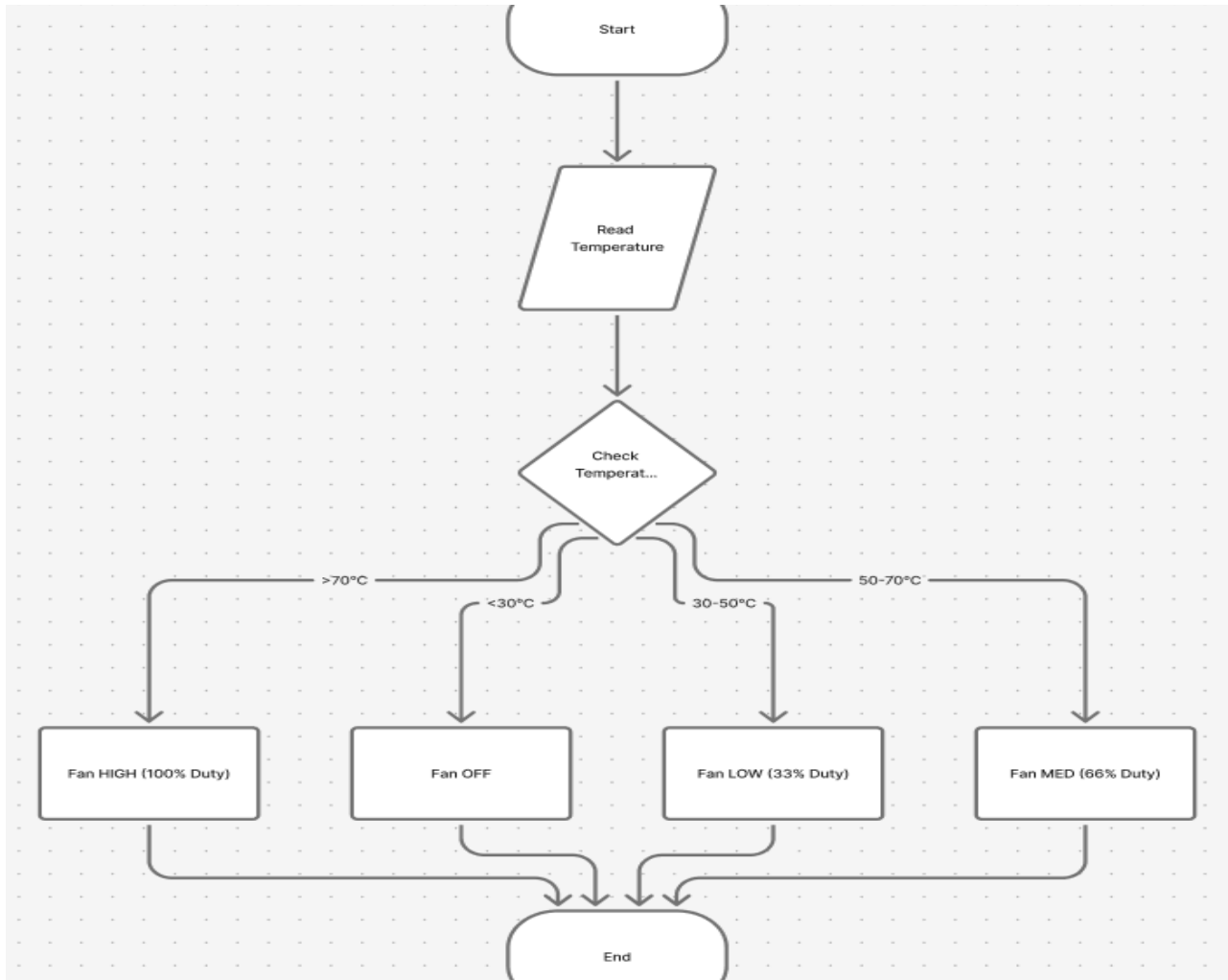
    // Simple PWM generator
    always @(posedge clk) begin
        pwm_counter <= pwm_counter + 1;
        if (pwm_counter < duty_cycle)
            fan <= 1;
    end
end

```

```
else
    fan <= 0;
end
```

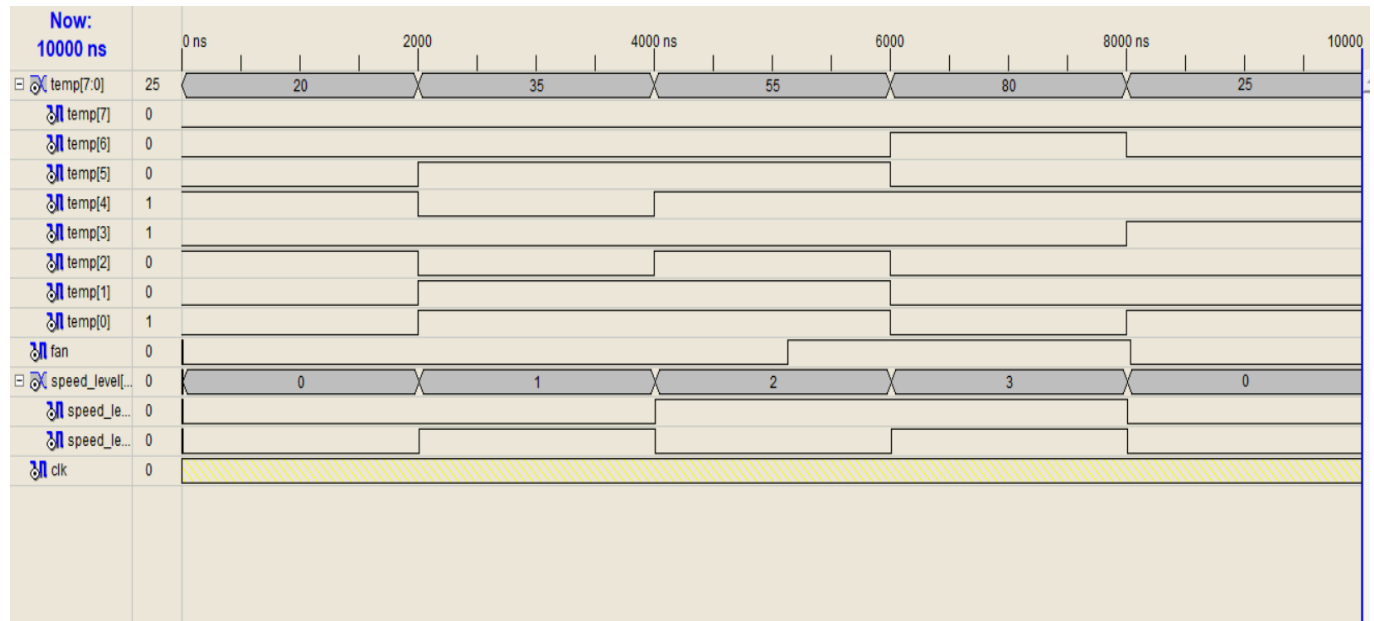
```
endmodule
```

b)Flowchart:



4)Results/Simulation:

Simulation was performed using Xilinx ISE 8.1



Summary:

TIME	TEMP	FAN STATUS	SPEED LEVEL
0	20	0	00[OFF]
2000	35	1	01[LOW]
2000	35	0	01
4000	55	1	10[MEDIUM]
6000	80	1	11[HIGH]
8000	25	0	00

Observations and Discussions : -

- When the temperature was below **30°C**, the fan remained **OFF**.
- For temperatures between **30°C and 50°C**, the fan ran at **low speed** with a small duty cycle.
- For temperatures between **50°C and 70°C**, the fan speed **increased** (medium speed).
- Above **70°C**, the fan operated at **full speed** with a 100% duty cycle.

- The **waveform** clearly showed an increase in PWM **duty cycle** as temperature rose.
- The system responded **accurately and smoothly** to temperature changes during simulation.

Conclusion and Learning outcomes : -

- The temperature-controlled fan system was **successfully designed and simulated** using Verilog on Xilinx.
- The fan speed was effectively **regulated using PWM** based on temperature changes.
- The system showed **accurate and smooth control** of fan speed in different temperature ranges.
- Learned how to **interface sensors and control outputs** through Verilog HDL.
- Gained experience in **simulating and analyzing waveforms** using Xilinx tools.
- Observed the **relationship between temperature and duty cycle** in a control system.

Faculty Name and Signature